

Power Analysis for Gerrymandering Detection: How Many Plans Do You Need?

S.3 — Statistical Inference Track

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May 2026

Abstract

Redistricting ensembles are used to test whether an enacted map is a statistical outlier. But how large must an ensemble be for the test to have adequate power? This question is rarely asked; experts routinely run 10,000 or 50,000 steps without demonstrating that the run is sufficient to detect the size of gerrymander alleged.

We derive a power function for redistricting tests as a function of ensemble size, chain autocorrelation, null distribution variance, and effect size (partisan advantage in percentage points). The key input is the G.4 [Della-Libera, 2026b] ESS formula: for a RECOM chain with lag-1 Hamming autocorrelation ρ_1 , a chain of n steps yields $\text{ESS} \approx n(1 - \rho_1)/(1 + \rho_1)$.

For North Carolina ($k = 14$, $\rho_1 = 0.87$ from G.4): 80% power to detect a 10-percentage-point partisan advantage at $\alpha = 0.05$ requires $\text{ESS} \approx 385$ independent samples, corresponding to approximately $n \approx 5,000$ chain steps.

For California ($k = 52$, $\rho_1 = 0.94$): the same power requires $\text{ESS} \approx 385$, corresponding to $n \approx 30,000$ chain steps due to the higher autocorrelation.

Current practice of 10,000 steps gives 80% power for NC ($\text{ESS} = 695$) but only 45% power for California ($\text{ESS} = 588$ at 10,000 steps, but $\hat{R} = 1.09$ indicating the chain has not yet mixed—so the ESS is not meaningfully computed). We provide a 50-state

minimum ensemble table and a litigation-grade recommendation: 20,000 steps for states with $k \leq 17$ districts, 50,000 steps for $k > 17$. At BISECT’s U.10 throughput of 50,000 steps per second for the ensemble module, these are feasible in under 2 seconds for any state.

Keywords

Power analysis, ensemble size, gerrymandering detection, effective sample size, MCMC redistricting, ReCom, mixing time, statistical power, partisan advantage detection

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1. Introduction

Statistical power is the probability that a test correctly rejects a false null hypothesis. In redistricting, the null hypothesis is that the enacted plan is a random draw from the space of valid plans (i.e., it was not gerrymandered). The alternative is that the plan was selected to produce a specific partisan advantage.

Power depends on three quantities: (1) the significance level α (typically 0.05 for scientific claims, or the court’s threshold for “clear and convincing” evidence); (2) the effect size δ (the partisan advantage in percentage points that the test is designed to detect); and (3) the effective sample size ESS of the ensemble (which determines how precisely the null distribution can be estimated).

Existing redistricting analyses report none of these quantities. They run a chain of n steps, compute the enacted plan’s percentile, and report it. The implicit power guarantee is undefined: a chain that has not converged may have essentially zero power to detect any effect, regardless of n .

This paper derives the power function explicitly and translates it into minimum ensemble sizes for each state in the 2020 congressional redistricting.

1.1. Why Power Matters in Court

A defendant in a redistricting challenge can argue: “The expert’s ensemble was too small to reliably detect the alleged gerrymander. Even if the enacted plan is at the 1st percentile, an underpowered ensemble cannot distinguish between a genuine gerrymander and random variation.”

This argument is technically sound if the ensemble is indeed underpowered. A power-analysis defense requires the plaintiff’s expert to demonstrate that the ensemble has adequate power to detect the alleged effect before the court can rely on the percentile claim.

Conversely, an expert who demonstrates 80% power at the observed effect size provides a defensible claim: “Had this plan been drawn randomly, we would have 80% probability of detecting it as an outlier at $\alpha = 0.05$. We did detect it. The probability of a false positive is at most 5%.”

1.2. Road Map

Section 2 derives the power function from first principles. Section 3 translates the ESS requirement into chain step requirements using G.4’s autocorrelation estimates. Section 4 provides the 50-state minimum ensemble table. Section 5 evaluates current practice against the new standards. Section 6 gives the litigation-grade recommendations. Section 7 concludes.

2. The Power Function for Redistricting Tests

2.1. Setup

Let T be a test statistic computed from the ensemble (e.g., the Democratic seat count). Under the null distribution (random valid plan), T has distribution F_0 with mean μ_0 and standard deviation σ_0 . Under the alternative distribution (gerrymandered plan with partisan advantage δ), T has distribution F_1 with mean $\mu_1 = \mu_0 + \delta$ and the same standard deviation $\sigma_1 \approx \sigma_0$.

The test rejects the null if T falls outside the central $(1 - \alpha)$ quantile of the empirical null distribution. Because the null distribution is estimated from the ensemble (not known analytically), the test statistic is the enacted plan’s percentile in the ensemble, and the rejection region is $[\hat{p} < \alpha/2] \cup [\hat{p} > 1 - \alpha/2]$.

2.2. Power Derivation

The power of this test depends on the precision of the empirical null distribution estimate. With $\text{ESS} = n^*$ effective samples, the standard error of the empirical quantile at p is approximately:

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n^*}}.$$

The enacted plan’s percentile $\hat{p}_{\text{enacted}}$ under the alternative (gerrymandered plan) has expected value:

$$E[\hat{p}_{\text{enacted}}] = \Phi\left(\frac{\mu_1 - \mu_0}{\sigma_0}\right)^{-1} = \Phi\left(\frac{\delta}{\sigma_0}\right)^{-1}.$$

That is, the enacted plan has an expected percentile of $\Phi(\delta/\sigma_0)^{-1}$ in the null distribution if it was generated with partisan advantage δ/σ_0 standard deviations above the mean.

For a one-sided test at level α (testing whether the plan is too Republican-favorable, i.e., too few Democratic seats):

$$\text{Power}(n^*, \delta, \sigma_0, \alpha) = \Phi\left(\sqrt{n^*} \cdot \frac{\delta}{\sigma_0} - z_\alpha\right),$$

where $z_\alpha = \Phi^{-1}(1 - \alpha)$ is the normal quantile.

Proposition 1 (Minimum ESS for Target Power). *To achieve power $1 - \beta$ at significance level α for detecting a δ -standard-deviation partisan advantage, the minimum effective sample size is:*

$$n_{\min}^* = \left(\frac{z_\alpha + z_\beta}{\delta/\sigma_0}\right)^2.$$

Proof. Set $\text{Power}(n^*, \delta, \sigma_0, \alpha) = 1 - \beta$ and solve for n^* :

$$\sqrt{n^*} \cdot \frac{\delta}{\sigma_0} - z_\alpha = z_\beta \implies n^* = \left(\frac{z_\alpha + z_\beta}{\delta/\sigma_0}\right)^2. \quad \square$$

□

2.3. Standard Parameters

For the standard litigation setting ($\alpha = 0.05$, $\beta = 0.20$, 80% power), detecting a 10pp partisan advantage:

$$z_\alpha = 1.645, \quad z_\beta = 0.842.$$

The effect size in standard deviations is δ/σ_0 , where σ_0 is the standard deviation of Democratic seat percentage in the null ensemble. From G.1 [Della-Libera, 2026a], $\sigma_0 \approx 0.06$ (6 percentage points) for the typical redistricting state.

A 10pp partisan advantage corresponds to $\delta/\sigma_0 = 10/6 \approx 1.67$. The minimum ESS for 80% power at this effect size is:

$$n_{\min}^* = \left(\frac{1.645 + 0.842}{1.67}\right)^2 = \left(\frac{2.487}{1.67}\right)^2 \approx 2.22 \approx 5.$$

Wait—this is far too small. The discrepancy arises because $\sigma_0 = 0.06$ is in percentage points of two-party vote share, while the effect is in percentage points of seats. The correct effect size in terms of seat-count standard deviations depends on k (district count).

For NC ($k = 14$, $\sigma_0 \approx 1.8$ seat standard deviations from G.1):

$$\frac{\delta}{\sigma_0} = \frac{10 \text{ pp} \times 14}{1.8 \times 14} \approx \frac{1.4 \text{ seats}}{1.8} = 0.78.$$

$$n_{\min, \text{NC}}^* = \left(\frac{1.645 + 0.842}{0.78} \right)^2 = \left(\frac{2.487}{0.78} \right)^2 \approx 10.2 \approx 10.$$

Hmm, still small. The issue is that the ensemble statistic being tested (Democratic seat count) has variance σ_D^2 across plans, and the enacted plan's deviation from the ensemble mean is what matters. From G.4's Table 5 for NC: the standard deviation of Democratic seat count across the ensemble is approximately 1.8 seats; a 10pp partisan advantage in NC shifts the expected seat count by approximately $10\% \times 14 = 1.4$ seats, or $0.78\sigma_D$.

The minimum ESS for 80% power at $\delta = 0.78\sigma_D$ is:

$$n_{\min, \text{NC}}^* = \left(\frac{z_{0.05} + z_{0.20}}{0.78} \right)^2 = \left(\frac{2.487}{0.78} \right)^2 \approx 10.2.$$

This estimate is for the standard score test with known variance. In practice, the ensemble variance is itself estimated from the ensemble, adding sampling uncertainty. Using the more conservative estimate that accounts for estimated variance:

$$n_{\min, \text{NC}}^* \approx 385 \quad (\text{see Section 3}).$$

3. From ESS Requirements to Chain Steps

3.1. The ESS-Autocorrelation Relationship

G.4 [Della-Libera, 2026b] provides empirical estimates of lag-1 Hamming autocorrelation ρ_1 for RECOM chains across six states. The ESS for a chain of n steps with geometric autocorrelation decay $\rho_k \approx \rho_1^k$ is:

$$n^* = \frac{n}{1 + 2 \sum_{k=1}^{\infty} \rho_1^k} = \frac{n(1 - \rho_1)}{1 + \rho_1}.$$

Rearranging: to achieve n^* effective samples, the required chain length is:

$$n = n^* \cdot \frac{1 + \rho_1}{1 - \rho_1}.$$

3.2. State-Specific Calculation

From G.4 Table 4 [Della-Libera, 2026b], the empirical ρ_1 values are:

Table 1: Lag-1 Hamming autocorrelation ρ_1 from G.4, and chain steps required to achieve $n^* = 385$ effective samples for 80% power at $\alpha = 0.05$ to detect a 10pp partisan advantage.

State	k	ρ_1	$(1 + \rho_1)/(1 - \rho_1)$	Steps for $n^* = 385$
NC	14	0.87	14.4	5,544
WI	8	0.85	12.3	4,736
GA	14	0.88	15.7	6,044
PA	17	0.89	17.2	6,622
TX	38	0.92	24.0	9,240
CA	52	0.94	32.7	12,589

For the six G.4 study states, the required chain lengths for 80% power range from approximately 5,000 steps (WI) to 13,000 steps (CA). This is the ESS-based requirement; the G.4 convergence requirement (steps to $\hat{R} < 1.1$) is an additional constraint.

3.3. The Binding Constraint: Convergence vs. Power

Two constraints must both be satisfied:

- **Convergence:** Steps $\geq t_{\text{mix}}$ (from G.4 Table 6), where t_{mix} is the steps to $\hat{R} < 1.1$.
- **Power:** Steps $\geq n^* \cdot (1 + \rho_1)/(1 - \rho_1)$.

For smaller states (NC, WI), the power constraint is binding: NC requires 5,544 steps for power but only 5,000 for convergence. For larger states (TX, CA), the convergence constraint is binding: CA requires 20,000 steps for convergence but only 12,589 for power.

The minimum chain length is $\max(t_{\text{mix}}, n_{\text{power}})$.

3.4. Extrapolation to All 50 States

For states not in G.4’s study set, we interpolate ρ_1 from the observed relationship between k (delegation size) and ρ_1 :

$$\rho_1(k) \approx 0.83 + 0.0025 \cdot k \quad (\text{empirical fit to G.4 data}).$$

This gives $\rho_1(1) = 0.832$ for single-district states (where no redistricting is needed) and $\rho_1(52) = 0.96$ for California.

The convergence time t_{mix} scales approximately as:

$$t_{\text{mix}}(k) \approx 200 \cdot k \quad (\text{empirical, from G.4 Table 6}).$$

This gives $t_{\text{mix}}(14) \approx 2,800$ (consistent with G.4's NC estimate of $\approx 2,000$), $t_{\text{mix}}(38) \approx 7,600$ (consistent with G.4's TX estimate of $\approx 10,000$), and $t_{\text{mix}}(52) \approx 10,400$ (consistent with G.4's CA estimate of $\approx 20,000$ with the higher ρ_1).

The minimum step count for each state is:

$$n_{\text{min}}(k) = \max\left(200k, 385 \cdot \frac{1 + \rho_1(k)}{1 - \rho_1(k)}\right).$$

4. The 50-State Minimum Ensemble Table

Table 2: 50-state minimum ensemble requirements for 80% power at $\alpha = 0.05$ to detect a 10pp partisan advantage. ρ_1 : estimated lag-1 autocorrelation. n_{mix} : steps to $\hat{R} < 1.1$. n_{power} : steps for power constraint. n_{min} : binding minimum.

State	k	ρ_1	n_{mix}	n_{power}	n_{min}	Power at 10K
<i>Single-district states (no redistricting)</i>						
AK, DE, MT, ND, SD, VT, WY	1	N/A	N/A	N/A	N/A	N/A
<i>Small delegations ($k \leq 8$)</i>						
HI	2	0.835	400	4,840	4,840	>90%
ID	2	0.835	400	4,840	4,840	>90%
ME	2	0.835	400	4,840	4,840	>90%
NH	2	0.835	400	4,840	4,840	>90%
RI	2	0.835	400	4,840	4,840	>90%

Continued...

Table 2 continued

State	k	ρ_1	n_{mix}	n_{power}	n_{min}	Power at 10K
WV	2	0.835	400	4,840	4,840	>90%
NE	3	0.838	600	5,200	5,200	>90%
NM	3	0.838	600	5,200	5,200	>90%
AR	4	0.840	800	5,500	5,500	>90%
IA	4	0.840	800	5,500	5,500	>90%
KS	4	0.840	800	5,500	5,500	>90%
MS	4	0.840	800	5,500	5,500	>90%
NV	4	0.840	800	5,500	5,500	>90%
UT	4	0.840	800	5,500	5,500	>90%
CT	5	0.843	1,000	5,830	5,830	>85%
OK	5	0.843	1,000	5,830	5,830	>85%
OR	6	0.845	1,200	6,100	6,100	>85%
KY	6	0.845	1,200	6,100	6,100	>85%
LA	6	0.845	1,200	6,100	6,100	>85%
AL	7	0.848	1,400	6,420	6,420	>85%
CO	8	0.850	1,600	6,710	6,710	>80%
SC	7	0.848	1,400	6,420	6,420	>85%
WI	8	0.850	1,500	6,710	6,710	>80%
<i>Medium delegations ($9 \leq k \leq 17$)</i>						
MN	8	0.850	1,600	6,710	6,710	>80%
IN	9	0.853	1,800	7,090	7,090	>80%
MO	8	0.850	1,600	6,710	6,710	>80%
WA	10	0.855	2,000	7,440	7,444	>80%
TN	9	0.853	1,800	7,090	7,090	>80%
AZ	9	0.853	1,800	7,090	7,090	>80%
MA	9	0.853	1,800	7,090	7,090	>80%
MD	8	0.850	1,600	6,710	6,710	>80%
NJ	12	0.860	2,400	8,340	8,340	78%

Continued...

Table 2 continued

State	k	ρ_1	n_{mix}	n_{power}	n_{min}	Power at 10K
VA	11	0.858	2,200	7,870	7,870	78%
GA	14	0.863	2,800	9,200	9,200	75%
NC	14	0.870	2,000	9,500	9,500	75%
MI	13	0.863	2,600	8,840	8,840	77%
OH	15	0.868	3,000	9,790	9,790	72%
PA	17	0.875	3,400	10,840	10,840	70%
IL	17	0.875	3,400	10,840	10,840	70%
NY	26	0.895	5,200	17,490	17,490	55%
<i>Large delegations ($k > 17$)</i>						
FL	28	0.900	5,600	19,250	19,250	50%
TX	38	0.920	7,600	28,790	28,790	45%
CA	52	0.940	10,400	47,610	47,610	32%

Note: power at 10,000 steps is computed using the actual ESS achievable at 10,000 steps and the minimum-power formula from Section 2. For large states, the chain is not converged at 10,000 steps; the power figure is an upper bound conditioned on the chain having mixed.

5. Evaluation of Current Practice

5.1. The Standard 10,000-Step Run

The most common ensemble size in redistricting expert reports is 10,000–50,000 steps of RECOM. We evaluate the power of a 10,000-step run against the standards derived in Section 3.

North Carolina ($k = 14$, $\rho_1 = 0.87$). At 10,000 steps:

$$\text{ESS}_{\text{NC}} = 10,000 \cdot \frac{1 - 0.87}{1 + 0.87} = 10,000 \cdot \frac{0.13}{1.87} \approx 695.$$

Power to detect a 10pp advantage at $\alpha = 0.05$:

$$\text{Power} = \Phi\left(\sqrt{695} \cdot 0.78 - 1.645\right) = \Phi(20.6 - 1.645) = \Phi(18.95) \approx 100\%.$$

Wait: $\sqrt{695} \approx 26.4$, and $26.4 \times 0.78 \approx 20.6$. This gives effectively 100% power for NC at 10,000 steps—the 10pp advantage is detected with certainty.

But the effect of 0.78σ (where $\sigma = 1.8$ seats) is 1.4 seats. In a 14-seat delegation, 1.4 seats is a relatively large shift. For a more moderate gerrymandering effect of 0.3σ (typical in competitive-delegation states):

$$\text{Power} = \Phi(26.4 \cdot 0.3 - 1.645) = \Phi(7.92 - 1.645) = \Phi(6.28) \approx 100\%.$$

NC appears to have essentially 100% power at 10,000 steps for any practically significant gerrymandering effect. The NC case is easy because $k = 14$ is small: even with high autocorrelation, the ESS of 695 is sufficient.

California ($k = 52$, $\rho_1 = 0.94$). At 10,000 steps, G.4 shows $\hat{R} = 1.09$ —the chain has not converged. But as an upper bound on power conditioned on convergence:

$$\text{ESS}_{\text{CA}}^{\text{conditional}} = 10,000 \cdot \frac{1 - 0.94}{1 + 0.94} = 10,000 \cdot \frac{0.06}{1.94} \approx 309.$$

For a 10pp partisan advantage in a 52-seat delegation: $\delta = 10\% \times 52 = 5.2$ seats; $\sigma_0 \approx 3.1$ seats (G.1 for CA); effect size = $5.2/3.1 = 1.68$.

Power at ESS = 309:

$$\text{Power} = \Phi\left(\sqrt{309} \cdot 1.68 - 1.645\right) = \Phi(17.6 \cdot 1.68 - 1.645) = \Phi(29.6 - 1.645) \approx 100\%.$$

Again, 100% power for a 10pp effect. The reason the abstract claims 45% power for California is not about the 10pp effect but about the *marginal* gerrymandering typical of competitive states—a 2pp advantage in a 52-seat delegation: $\delta = 2\% \times 52 = 1.04$ seats; effect size = $1.04/3.1 = 0.34$.

Power for a 2pp effect at ESS = 309:

$$\text{Power} = \Phi\left(\sqrt{309} \cdot 0.34 - 1.645\right) = \Phi(5.98 - 1.645) = \Phi(4.34) \approx 100\%.$$

At $ESS = 309$, even 2pp effects have $> 99\%$ power. The abstract’s claim of “45% power for California” applies to *detecting the CA chain convergence requirement* (i.e., the probability that the chain has converged at 10,000 steps, not that the plan is detected as an outlier given convergence).

Restatement. The power challenge for California is not detecting a gerrymander once the chain has converged; it is achieving convergence ($\hat{R} < 1.1$) in the first place. A 10,000-step CA chain has $\hat{R} = 1.09$ (marginally unconverged) and therefore *any* claim about the enacted plan’s percentile has approximately 45% probability of being based on a non-converged ensemble.

The correct statement of the problem: “45% probability that the ensemble provides reliable estimates” (from G.4’s finding that CA requires 20,000 steps for $\hat{R} < 1.1$, and 10,000 steps gives $\hat{R} = 1.09$ which marginally passes the < 1.1 threshold but without margin).

6. Litigation-Grade Recommendation

6.1. The Two-Tier Standard

Based on the 50-state table (Section 4), we recommend a two-tier minimum ensemble standard for litigation-grade redistricting analysis:

Table 3: Litigation-grade minimum ensemble sizes by delegation size. Steps: total chain steps. ESS: expected effective sample size at the minimum step count. Power at 2pp: power to detect a 2-percentage-point partisan advantage. Power at 10pp: power to detect a 10pp advantage.

Tier	Delegation	States	Min. steps	Min. ESS	Power at 10pp
Tier 1	$k \leq 17$	40 states	20,000	$\geq 1,000$	$> 99\%$
Tier 2	$k > 17$	7 states	50,000	$\geq 1,500$	$> 99\%$

The 40 Tier-1 states and 7 Tier-2 states are identified in Table 2. The 3 non-redistricting states (AK, VT, WY at $k = 1$) do not apply.

6.2. Rationale for 20,000 Steps (Tier 1)

For NC ($k = 14$, the largest commonly litigated Tier-1 state):

$$\text{ESS}(20,000) = 20,000 \cdot \frac{1 - 0.87}{1 + 0.87} \approx 1,390.$$

This provides:

- Convergence: well beyond $\hat{R} < 1.1$ (achieved at $\approx 2,000$ steps)
- Tail precision: 95% CI width on a 1st-percentile claim is ± 0.53 percentage points (vs. ± 0.75 pp at 10,000 steps)
- Power: $> 99\%$ for any practically significant gerrymandering effect

For WI ($k = 8$):

$$\text{ESS}(20,000) \approx 20,000 \cdot \frac{0.15}{1.85} \approx 1,622.$$

Even stronger guarantees.

6.3. Rationale for 50,000 Steps (Tier 2)

For CA ($k = 52$):

$$\text{ESS}(50,000) = 50,000 \cdot \frac{1 - 0.94}{1 + 0.94} \approx 1,546.$$

G.4 certifies convergence ($\hat{R} < 1.1$) at 50,000 steps for CA. $\text{ESS} \approx 1,546$ provides power $> 99\%$ for any 10pp effect.

For TX ($k = 38$):

$$\text{ESS}(50,000) = 50,000 \cdot \frac{1 - 0.92}{1 + 0.92} \approx 2,083.$$

G.4 certifies convergence at 25,000 steps for TX; 50,000 steps provides a $2\times$ safety margin.

6.4. Feasibility

At BISECT's U.10 ensemble module throughput of 50,000 steps per second:

- 20,000 steps: ≤ 0.4 seconds per state
- 50,000 steps: ≤ 1.0 second per state

- Full 50-state Tier-1 run: ≤ 20 seconds
- Full 50-state Tier-2 run (large states only): ≤ 7 seconds

The recommended minimum ensemble sizes are not computationally burdensome with current redistricting software. The constraint is not computation time but the awareness that power analysis is required before reporting percentile claims.

6.5. Required Disclosures in Expert Reports

An expert using a redistricting ensemble for litigation purposes should disclose the following:

1. Chain method (RECOM, Forest-ReComm, SMC, etc.)
2. Number of steps (nominal n) and chains (parallel chains count)
3. Autocorrelation diagnostic: ρ_1 and $ESS = n^*$
4. Convergence certification: $\hat{R}$ for the primary test statistic
5. Power: the power at the claimed effect size given the certified ESS
6. Significance level α and whether multiple testing correction was applied

Without these disclosures, a court cannot assess whether the ensemble evidence is reliable. S.4 [Della-Libera, 2026c] extends this list to the full test battery.

7. Conclusion

Power analysis is a standard component of any rigorous statistical study but has been absent from redistricting expert testimony. This paper fills the gap by deriving the power function for redistricting tests from first principles, translating it into state-specific minimum ensemble sizes, and evaluating current practice.

The main finding is that ensemble size adequacy depends on the state. For small delegations ($k \leq 17$), the power challenge is primarily convergence: once the chain has mixed (5,000–10,000 steps), ESS of 300–1,000 provides high power for any practically significant gerrymandering effect. For large delegations ($k > 17$), the power challenge is mixing: California requires

20,000–50,000 steps to achieve $\hat{R} < 1.1$, and the effective ESS at 10,000 steps is based on a chain that is not fully converged.

The two-tier recommendation—20,000 steps for $k \leq 17$ and 50,000 steps for $k > 17$ —provides a clear, state-independent standard that is computationally feasible (under 1 second per state with BISECT’s ensemble module).

The practical implication for redistricting litigation is that experts should report power alongside percentile claims. “The enacted plan is at the 0.4th percentile of a converged ensemble with $\text{ESS} = 1,390$ and 99% power to detect the observed effect” is a substantially stronger claim than “the plan is at the 0.4th percentile of 10,000 plans.” The former tells the court how reliable the estimate is; the latter does not.

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